

WILL MCGOWAN

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EDUCATION

University of Strathclyde, Glasgow

Sep 2023 - Jun 2027 (expected)

BA (Hons) Economics

- Average grade to date: 2:1
- **Relevant Modules:** Applied Econometrics; Treasury Management and Derivatives; Advanced Microeconomics

Dissertation | MATLAB

In progress, due 2027

"Does the News Pay? Forecasting European Gas Volatility with MIDAS, Storage and Text Data"

- Testing whether news attention improves forecasts of TTF gas price volatility with mixed-frequency (MIDAS) models built on daily storage, market and news data, including an original Google Trends attention index
- Benchmarking out-of-sample against HAR and AR models with Diebold-Mariano tests across pre-crisis, crisis and post-crisis periods, and measuring economic value through the Sharpe ratio of a volatility-targeted TTF futures strategy

Empirical CAPM Test

MATLAB, Python

- Tested CAPM on 704 anomaly portfolios across 56 countries (JKP global factor data), regressing each on its local market excess return
- Rejected zero alpha for 32% of portfolios at 5% (32.7% with Newey-West errors) and jointly via GRS in 48 of 52 countries; momentum alphas were significant in 37 of 55 countries, size in only 4 of 56

Monte Carlo Option Pricing Engine

MATLAB, Python

- Built a vectorised Monte Carlo engine pricing European and Asian options under risk-neutral GBM, benchmarked against Black-Scholes and the closed-form geometric-Asian price
- Reduced the standard error of arithmetic-Asian estimates by 97.24% vs plain Monte Carlo at $N = 100,000$ using a geometric-Asian control variate, alongside antithetic variates

Corn-Soybean Cointegration | group project

R

- Found one cointegrating relationship between corn and soybean prices (quarterly, 1990-2022) with Engle-Granger and Johansen tests; the error-correction model showed both prices adjusting back to equilibrium (soybean coefficient -0.29, $p < 0.001$)

Belfast Royal Academy, Belfast

Jun 2023

- A-levels: Mathematics (A), Economics (A), Computer Science (B)

WORK EXPERIENCE

Le Set, Glasgow | Head Mixologist

Sep 2026 - Present

- Created 4 original house cocktails, wrote the specifications for a new 200-cover venue's 11-drink menu and trained around 15 bar staff to deliver it

KONG, Glasgow | Mixologist and Front of House

Oct 2024 - Jun 2026

- Analysed sales patterns to refine upselling prompts, raising average customer spend by an estimated 25%; reconciled cash and card takings at the end of each shift

LEADERSHIP AND ACTIVITIES

Strathclyde Agricapital Society | Co-Founder and Treasurer

Sep 2025 - Present

- Co-founded the university's first agri-finance society, grew it to 126 members and published weekly research notes
- Built a wheat calendar-spread strategy in R from futures curve structure and contrarian CFTC managed-money positioning, with all parameters fixed before testing
- Weekly backtest (Apr 2025 - Oct 2026): Sharpe 0.76 before costs and 0.28 after, maximum drawdown 3.1%; identified turnover of about 50x notional a year as the main drag on returns

Strathclyde Poker Society | Vice-President, then President

Sep 2025 - Apr 2026

- Ran weekly tournaments for around 60 players on a £600-per-semester budget, reconciling every buy-in and prize pool

ADDITIONAL INFORMATION

Technical: R, MATLAB, Python, Excel, Git

Methods: time-series econometrics (cointegration, error correction, HAR, MIDAS), asset-pricing tests (OLS and Newey-West alphas, GRS), Monte Carlo with variance reduction, backtesting with costs and volatility targeting

Interests: poker, over £40,000 in lifetime cash-game winnings under a 50-buy-in bankroll rule; chess, 14th at the Irish U19 Chess Championships (2023) and won 21 of 22 games in a simultaneous exhibition against players averaging 1860; powerlifting, Northern Ireland U19 champion (2023)